

CPUs, GPUs, TPUs

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CPU_s

Each CPU consists of:

- ▶ one or more Arithmetic Logic Units (ALU) which performs arithmetic and logic operations
- ▶ the memory of the CPU, called *registers* or L1 cache
- ▶ a Control Unit (CU) which tells the other components how to execute instructions.

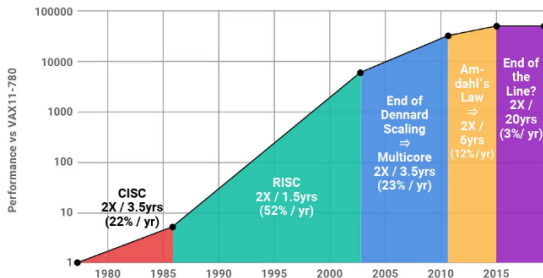
Advantages and disadvantages:

- ▶ + A CPU is extremely flexible
- ▶ - Must store the calculation results on the registers for every calculation. This gives rise to the so-called *von Neumann bottleneck*: each ALU executes one operation at a time.

Moore's Law

Moore's Law: # of transistors in CPUs used to double every 1.5 y.

40 years of Processor Performance



Source: cloud.google.com

How to cope with the end of growth? Specialized hardware...

GPUs

First GPU dates back to 1999 (NVidia GeForce 256) built for the game industry. Nowadays, used also for machine learning (especially deep learning) and bitcoin mining.

Each GPU contains 2,500 – 5000 ALUs $\Rightarrow$ much higher degree of parallelism than CPUs. They have much more internal memory than CPUs (e.g. 12GB in NVIDIA Titan Xp).

However, they still suffer from the *von Neumann bottleneck*.

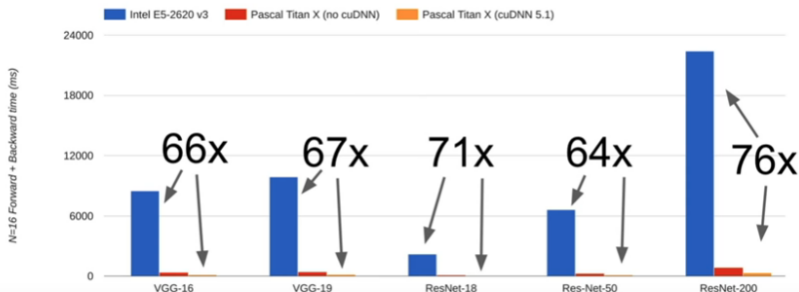


Figure: A GPU. It has its own memory and cooling system.

CPU vs GPU

CPU vs GPU in practice

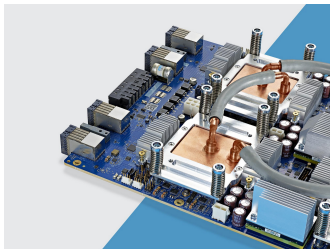
(CPU performance not well-optimized, a little unfair)



Data from <https://github.com/jcjohnson/cnn-benchmarks>

TPUs

Domain-specific architectures focused on tensor processing, developed at Google. Thousands of multipliers and adders are connected with each other to form a large physical matrix, called *systolic array architecture*.



- ▶ + No need to store intermediary results in main memory
- ▶ + higher throughput per watt
- ▶ - specialized, not suitable for other tasks

TPUs: Misc

- ▶ Announced in May 2016 by Google, but used for 1+ year.
- ▶ Designed specifically for Google's TensorFlow.
- ▶ Google TPUs are proprietary, however, accessible to other companies through its cloud-computing service.
- ▶ Used in the AlphaGo-versus-Lee-Sedol Go games.
- ▶ Used for Google Street View text processing.
- ▶ One single TPU can train on 100M images from Google photos per day.